

Indian Institute of Engineering Science and Technology, Shibpur

BTech/BArch 7th Semester Mid-Semester Examination, September, 2025

Finance, Economics and Management for Engineers (HU-4101)

Full Marks: 30 ;

Time: Two Hours

Answer Module wise. Start each module from a separate Page.

Finance Module(10 Marks)

- A. Prepare a Cost Statement in prescribed Format showing each stage with imaginary items. (4)
- B. Write notes (any Two):
- i. Fixed assets ii. Internal Liabilities iii. Process Costing iv. Elementwise classification of Cost v. Measurement in Accounting. (2x3)

Module: II; Economics(10 marks)

1. Why is magnitude of money flow constant in a circular flow of income model? Why do you think national income and national expenditure are identical in the model? (1+2)
 2. Why do you think the Keynesian cross is a stable equilibrium, if at all? What is an expansionary fiscal policy? (1+2)
 3. State the functions of money. Define inflation and state its causes. (2+2)
- OR
4. What is law of demand? Show how equilibrium price is determined in a market using demand and supply curves. (1+3)

Management Module(10 Marks)

Case Study: Infosys and the Evolution of Management Style

In its early years, Infosys operated with a relatively autocratic management style. Decisions were centralized among the founders, which helped the company maintain strict control, speed in execution, and consistent quality during its growth phase. However, as the company expanded globally, this rigid style began to show limitations—employee morale and innovation suffered because decision-making was too top-heavy.

In response, Infosys gradually adopted a democratic and participative style of management. Leaders began to involve employees more in project-level decisions, encouraged feedback sessions, and introduced open-door policies. This shift improved collaboration, built trust, and made Infosys a more attractive place to work.

At the same time, leaders like Narayana Murthy displayed transformational Leadership — emphasizing values, ethics, and a vision of creating a globally respected Indian IT company. By inspiring employees to look beyond short-term targets and align with the company's mission, Infosys built a culture of ownership and excellence.

This journey shows how an organization can adapt its management style over time to remain competitive, innovative, and people-focused.

Attempt any two questions

1. Why was an autocratic management style useful in the early stages of Infosys, and why did it become a limitation later? (5)
2. As a manager, how would you balance democratic participation with the need for quick decision-making in a fast-paced IT environment? (5)
3. Do you think transformational leadership is more sustainable than autocratic leadership in knowledge-based industries like IT? Why or why not? (5)

(Use separate answer scripts for each half)

First Half

(One Mark for neatness and to the point answers)

Finance Module(16 Marks)

1. An Equipment A has a cost of ((₹) 75,000/.

The net cash inflows over the estimated life of the Equipment are as follows :

1st year ((₹) 20000/,

2nd year ((₹) 18,000/

3rd year ((₹) 22,000/

4th year ((₹) 25,000 and

5th year ((₹) 23,000/ .

Another Equipment B has a cost of ((₹) 50,000/

The net cash inflows over the estimated life of the Equipment are as follows :

1st year ((₹) 14,000/

2nd year ((₹) 18,000/

3rd year ((₹) 12,000/

4th year ((₹) and 13,000

5th year ((₹).11,000/ .

The required return for both the equipment is 11% .

Calculate The NPV for both the projects and Comment on the acceptability.

(12)

OR

Suppose, ABC Ltd. which is expecting the EBIT of Rs.1,50,000 per annum on an investment Rs.5,00,000, is considering the finalization of the capital structure or the financial plan. The company has access to raise funds of varying amounts by issuing equity share capital, and 10% debenture or any combination thereof. Suppose, it analyses the four options to raise the required funds of Rs.5,00,000 which are as follows :

1. By issuing equity share capital at par.
2. 50% funds by equity share capital and 50% funds by issuing Debentures.
3. 75% funds by equity share capital, and 25% by issue of 10% debentures.
4. 25% funds by equity share capital, and 75% by the issue of 10% debentures.

Assuming that ABC Ltd. Has a 50% tax bracket, calculate the EPS under the above four options (12)

2. Write Notes (any one)

- a. Cost
- b. Cost of Capital.
- c. PBP
- d. Debt Capital.

(4)

Second Half

(One Mark for neatness and to the point answers)(Use Separate Answer scripts)

Module 2

Economics(16 Marks)

1. (a) Calculate GDP and GNP at market prices from the data given below (2+2)

<u>Expenditure</u>	<u>Amount (in Rs. Crore)</u>
Consumption	800
Investment	500
Factor payments to abroad	200
Factor income from abroad	50
Government expenditure	10
Exports	200
Imports	100

(b) What do you understand by real GDP? What advantages does it have over nominal GDP? Mention one flaw of using GDP as a measure of welfare. (3)

2. Explain the concept of marginal utility. Given the law of diminishing marginal utility and the first-order condition of utility maximisation, how much of a good X the consumer will consume if he is provided X free of cost? Explain in detail with the help of a utility diagram. (3)

OR

Obtain the utility maximising quantities of commodity x and commodity y that a consumer will consume where his utility function and budget line is represented below,

Utility: $U = x^{1/2} y^{1/2}$; $2x + 3y = 10$
 Budget constraint: where all symbols have usual meanings.

(3)

3. What is the difference between the short-run and long-run production function?

In drawing a Total Product (TP) curve, what is the significance of the inflexion point or the point at which the TP curve changes curvature? What law governs the shape of the TP curve? (3)

4. In what way are inflation and unemployment related in the short-run? Do episodes of high inflation and high unemployment negate the validity of the Phillips curve? Explain. (1+2)

Module 3

Management (16 Marks)

Part A: Answer any one case study (10 marks)

Case Study 1: The Strategic Reinvention of Nokia

Once the undisputed leader of the global mobile phone industry, Nokia lost its competitive edge during the smartphone revolution. Its failure to anticipate the dominance of iOS and Android, combined with organizational rigidity and late technological adaptation, resulted in a steep decline in market share. In 2013, Nokia sold its mobile handset business to Microsoft—a move initially seen as an exit from its core market. However, the company redefined its corporate identity by realigning toward network infrastructure, digital connectivity, and technology licensing.

In the following years, Nokia acquired Alcatel-Lucent to strengthen its position in network solutions, invested heavily in 5G research, and divested non-core operations to streamline its portfolio. Despite these shifts, it continues to face fierce competition from players such as Huawei, Ericsson, and Samsung in the telecom equipment market. The company's strategic dilemma now lies in balancing its focus between technological innovation and financial sustainability, while responding to global supply chain challenges and evolving market demands.

Questions:

- Evaluate Nokia's transformation using the lenses of retrenchment, turnaround, and renewal strategies. To what extent did these align with the firm's redefined corporate mission? (5)
- Suggest a strategic combination approach for Nokia to sustain competitiveness in the 5G and AI-driven ecosystem. Should the company pursue vertical integration, strategic alliances, or diversification? Substantiate your recommendation with analytical reasoning. (5)

OR

Case Study 2: Reliance Industries:

From Petrochemicals to Platform Power (10 Marks)

Reliance Industries Ltd. (RIL) has undergone one of the most ambitious corporate transformations in India's business history. Traditionally known for its dominance in petrochemicals and refining, RIL diversified into telecommunications (Jio) and retail in the early 2000s, building two of India's largest consumer-facing businesses within a decade. By 2020, RIL's digital ecosystem and retail network had become key drivers of growth, significantly contributing to profitability.

However, the company's strategic shift also involved calibrated retrenchment—divesting several non-core assets in oil and gas exploration, and reducing capital exposure in slower-growth segments. Simultaneously, RIL made aggressive investments in renewable energy, including solar, hydrogen, and energy storage technologies, with the goal of becoming net carbon-zero by 2035. This simultaneous pursuit of growth, retrenchment, and green transformation reflects an intricate combination strategy, positioning RIL as both a traditional industrial leader and a digital-technology conglomerate.

Questions:

a) Using Ansoff's Product–Market Growth Matrix, analyze how Reliance's diversification into telecom, retail, and renewable energy has expanded its strategic scope while leveraging existing competencies. (5)

b) Critically assess whether RIL's selective divestment in fossil-based businesses represents strategic flexibility or a risk mitigation measure in response to global sustainability trends. What challenges might arise in managing such a diversified corporate portfolio? (5)

Part B: Answer any two questions (3X2=6 marks)

1. Strategic Choice under Market Saturation:

During an economic slowdown, a consumer electronics firm chooses to maintain its existing product lines and focus on operational efficiency rather than launching new products. Critically examine how this reflects a stability strategy, and justify whether it is strategically sound in the long run.

2. Simultaneous Growth and Retrenchment:

Suppose an automobile manufacturer closes one of its underperforming plants while simultaneously investing in electric vehicle production. Evaluate this move as a combination strategy, discussing how it reflects both retrenchment and growth dimensions.

3. Integration Decision under Competitive Pressure:

A mid-sized FMCG firm faces rising input costs and increasing competition from new entrants. Analyse whether pursuing vertical integration (e.g., acquiring suppliers) or horizontal integration (e.g., acquiring competitors) would be more strategically advantageous. Justify your reasoning with reference to control, cost efficiency, and market power.

Finance Economics and Management for Engineers (HU4201)

Full Marks: 30

Time: 2 Hours

Module I: Finance

(Attempt any two)

1. Discuss AAA definition of Accounting. (5)
2. Write Short Notes on i. Contract Costing ii. Process Costing (5)
3. From the following information calculate the Raw Material Consumed (in INR) :

Op stock of Raw Materials	1,25,000	Carriage on Raw Materials purchased	500
Repairs to plant	9,600	Rent and rates of workshop	42,500
Raw Materials returned to suppliers	32,000	Non-productive Wages to workers	22,000
Fuel, gas, water etc.	21,000	Cl stock of Raw Materials	10,800
Productive Wages to workers	10,000	Depreciation on machinery	41,400
Office expenses	41,500	Direct chargeable expenses	2,800
Salaries paid to office staff	45,000	Purchase of Raw Materials	1,70,000

(5)

Module II: Economics

(Attempt all questions)

4. Define GDP of an economy. With the help of the Circular flow of income model, explain why total factor income is equivalent to total expenditure (2+3)
5. What do you understand by the marginal propensity to consume (mpc)? Given the range of values *mpc* can take, state its significance for the investment multiplier. Why is Keynesian Cross stated to be a stable equilibrium? (1+2+2)

Module III: Management

(attempt section A or section B)

- A.** Answer the following questions in brief:

[5 x 2 = 10]

- i. Explain the role of corporate social responsibility (CSR). How can CSR activities address the SD goals?

- ii. Do you consider management a “science,” “art,” or “profession”? Why?
- iii. As an engineer, how should you systematically coordinate management functions?
- iv. Develop a model to reduce employee attrition rate.
- v. How can you reduce capacity leakage in an organization?

OR

B. Case Study: Navigating Growth and Employee Engagement at SwiftTech

Scenario: SwiftTech, a fast-growing software company, has been expanding its market share. Initially a startup, the company has now reached mid-sized status, employing over 500 people. However, its management practices have not evolved in line with its growth, leading to several operational challenges. The key challenges are: -

- a) **Rigid Management Structure:** Top management makes decisions, leaving middle and lower management with little say in key operational matters. Employees feel disengaged.
- b) **Productivity Issues:** Due to rapid expansion, coordination between teams is lacking. Some projects experience delays and cost overruns because of ineffective communication.
- c) **Employee Morale & Retention:** The best performers feel undervalued, and turnover rates have increased by 20% over the last year.
- d) **Process Improvement Needs:** While the company has great ideas, there is no system for continuous improvement, leading to inefficiencies in operations.

CEO Mr. Arjun wants to restructure the management approach and implement effective leadership strategies to improve decision-making, engagement, and long-term success.

Answer the following questions based on the above case study **[5 x 2 = 10]**

- i. Identify the management challenges SwiftTech is facing. Which principles of management (as per Henri Fayol) are missing in SwiftTech’s approach?
- ii. What type of leadership style should Mr. Arjun adopt to improve employee motivation? Explain with examples.
- iii. Compare top-down decision-making with a participative management approach. Which would be more effective in this situation?
- iv. How can SwiftTech use Toyota’s Lean Management principles to enhance efficiency and employee involvement?
- v. Propose a new management structure that includes top, middle, and lower-level managers for better coordination and decision-making.

FINANCE, ECONOMICS & MANAGEMENT for ENGINEERS(HSS III)(HU - 4201)

Full Marks: 30

Time: 2 Hours

(ANSWER MODULEWISE AND START WRITING ANSWERS
FOR EACH MODULE ON A SEPARATE PAGE)
Each Module carries 10 marks (3x10=30).

Module I (FINANCE)

- A. What do you understand by Cost? Explain with example.
OR Discuss cost classifications according to Traceability. (4)
B. From the following Particulars calculate only Works Overheads :

<u>Items</u>	<u>Rs</u>	
Direct Materials	10,00,000/	
Direct Wages	2,00,000/	
Indirect Materials	60,000/	
Indirect Wages	1,00,000/	
Advertisement	50,000/	
Rent:		
Office	1,15,000/	
Factory	75,0000/	
Lighting	88,0000/	
(50% for Factory)		
Power	60,000/	
Depreciation of Plant	1,15,500/	
Salary of Works Manager	95,000/	
Lubricant	10,000/	
Factory Guards Salary	30,000/	
Salary of Administrative staff	80,000/	
Insurance of Plant	25,000/	(6)

Module II

Economics

1. Read the paragraph and answer the questions that follow.

“If inflation were higher, critics would argue that nominal GDP growth is much higher because of inflation and that there was little underlying activity. MoSPI calculates quarterly GVA in real terms first, and then, using the deflator, nominal values are obtained. No wonder nominal growth rates have slowed, with WPI contracting in recent months. This will normalise in the coming months,” the statement pointed out.

Source- The Hindu Bureau, September 15, 2023

- (a) What do you understand by the term nominal GDP? How is it different from real GDP? (2)

(b) Define GDP deflator. (2)

(c) Why do you think the critics in the above paragraph would argue that higher nominal GDP growth along with higher inflation cast doubt on the real underlying economic activity? (3)

2. What was the Keynesian suggestion to uplift the economy from a period of sustained recession? If the investment multiplier has a value Five (5), calculate the value of marginal propensity to consume in the economy. (1+2)

OR.

State the functions of money. What do you understand by the term Broad Money? (1+2)

Module III

Management

1. Discuss in detail contributions of F.W Taylor and Henry Fayol in the Scientific Management era and Modern Management era respectively for developing the Management subject. (10)

Indian Institute of Engineering Science and Technology, Shibpur
B Tech/B Arch 8th Semester End-Semester Examination, April, 2024
FINANCE ECONOMICS & MANAGEMENT for ENGINEERS (HU - 4201)

Full Marks: 50

Time: 3 Hours

(Use Separate Answer scripts for each half. Answer Module wise and start writing Answers for each

Module on a separate page)

Each Module carries 16 marks (3x16=48) and 1 mark for neatness in each half.

1st Half

Finance Module (16)

1. Explain the with Examples (any two)

- a. Functional classification of Cost b. Pay Back Period c. Internal Rate of Return
b. d. Long Term Sources of Corporate Finance . (4x2=8)

2. From the following information calculate degree of Operating Leverage and Financial

Leverage and comment :

	<u>Situation I (Rs '000)</u>	<u>Situation II (Rs '000)</u>
Sales	100	150
Variable Cost	40	60
Fixed Cost	30	30
Total Cost	70	90
Profit	30	60
Interest	04	04
Tax	13	27
Net Surplus	13	27
No of Equity Shares	10	10

(4+4=8)

MANAGEMENT MODULE(16)

Answer any two questions

3. (a) What are the differences between the strategies of business level and corporate level?
(b) Explain cost leadership strategy with an example. [4+4= 8]
4. What are the main three types of management styles? Explain their characteristics with examples in brief. [1+7=8]
5. (a) How to measure managerial success ?
(b) What are the importance and needs of management ? [4+4= 8]
6. (a) Discuss in brief about different types of functional strategies ?
(b) What are the importance of communication in an organisation ? [5+3= 8]

2nd Half

Use separate Answer script

1. A consumer has a utility function of the form $U = x^{\frac{1}{2}}y^{\frac{1}{2}}$ and its budget is represented by the equation $10x + 20y = 200$, where the symbols have their usual meanings. Calculate at what point of consumption of *good x* and *good y* does the consumer maximizes its utility?
(6)
2. In the context of the theory of consumer behaviour discuss what is meant by Rationality of a consumer. If you vote for Party A in Lok Sabha election over Party B and then Party B over Party A in Assembly election, what Axiom of Choice are you violating, if any? Explain.
(3+2)
3. (a) Explain how fractional reserve banking creates money supply in an economy.

(b) Fill in the value of **x** and **y** in the following table:

Units	Total Product (TP)	Marginal Product (MP)	Average Product (AP)
1	10	-	10
2	20	10	10
3	35	x	11.67
4	45	10	y

(3+2)

OR

4. Explain diagrammatically how an autonomous increase in government expenditure affects equilibrium income in a Keynesian economy. State one important assumption of Keynes' model of income determination.

(4+1)

Indian Institute of Engineering Science and Technology, Shibpur
BTech/BArch 7th Semester Mid-Semester Examination, September, 2024

Finance Economics and Management for Engineers (HU4101)

Full Marks: 30

Time: 2 hours

Follows instructions of each module separately. Marks for each question are indicated in the brackets alongside the questions

Module I (Finance)

1. Discuss four major Long term Sources of Corporate Finance. (4)
2. a. What is Cost? Explain the Functional Classification of Cost with an example. (6)
OR
b. From the Following information calculate the Administrative Overheads only :

<u>Items</u>	<u>Amount (₹)</u>
Packing Charges	50,000
Clerk's Salary	1,80,000
Postage	5,000
Sales	90,00,000
Stationery	80,000
Indirect wages	50,000
Meeting Expenses	50,000
Direct Wages	8,00,000
Accounting Charges	2,50,000
Direct Expenses	2,00,000
Audit Fees	1,60,000
Advertisements	1,00,000
Office Rent	1,50,000
Depreciation of Photo Copier	50,000
Direct Materials	10,00,000
Electricity (50% for Factory)	1,00,000
Telephone charges	70,000

(6)

Module II (Economics)

Answer all questions

1.(a) What is GDP of a country? (1)

(b) Goods and services produced in the previous year will not be included in calculating the current year's GDP. Explain. (1)

(c) What can you infer about the nature of a country if its GDP is very low compared to its GNP? (1)

(d) How does inflation affect the calculation of an economy's real GDP? (2)

2. (a) When the monetary authority of a country decides to increase the money supply in an economy, reserve requirement rates are cut. Explain the (in)validity of the previous statement using the money multiplier mechanism. (3)

(b) State and explain the functions of money. (2)

Module III (Management)

Answer any one question

1. (a) Briefly elaborate on the various functions of management.

(b) What are the various challenges to management?

(2 x 5 = 10)

(c) Briefly explain the different levels of management.

(d) What is CSR? What are the benefits of CSR?

(e) What are the types of CSR?

2. (a) Google's Project Aristotle was a research initiative aimed at understanding what makes teams successful. What is the relevance of management in making successful teams? **(5)**

(b) Consider the various functions of management, such as planning, organizing, etc. In your opinion, how do these functions require both systematic, evidence-based approaches and intuitive, creative thinking? Can you identify areas where management relies more heavily on one over the other? Support your arguments with real-world examples. **(5)**

Indian Institute of Engineering Science and Technology, Shibpur
BTech/BArch 7th Semester End-Semester Examination, November, 2024

Finance Economics and Management for Engineers (HU4101)

Full Marks: 50

Time: 3 hours

Write Module I and II in 1st half while Module III in 2nd half. Follow the instructions of each module separately. Marks are indicated in parentheses alongside the questions.

1st Half

Module I: Finance (Marks 17)

1. According to Functional classification, indicate what type of Overheads are the following items (any Seven):

a. Lubricants b. Advertisement c. Carriage outwards d. Meeting Expenses e. Depreciation of Delivery Van f. Stationery g. Power and Fuel. h. Cotton Waste i. Director's Fees j. General Expenses. (7)

2. Explain the concept of Operating Leverage, Financial Leverage and Combined Leverage. (10)

OR

A company is considering whether to purchase a new machine. Machines A and B are available for Rs. 80,000 each. Earnings after taxation are given below.

Year	Machine A (Rs)	Machine B (Rs)
1	24,000	8,000
2	32,000	24,000
3	40,000	32,000
4	24,000	48,000
5	16,000	32,000

Required: Evaluate the two alternatives using the Net Present Value Method. You should use a discount rate of 10%. (10)

Module II: Economics (Marks - 17)

Answer all questions from 1 to 4 and any one between 5 and 6

1. State the law of demand. What assumption in the cardinal utility analysis ensures the slope of the demand to be the way it is? (1+2)
2. What do you understand by rationality of a consumer? If the axiom “averages are preferred to extremes” holds, what can you infer about the shape of the Indifference curve? Is the axiom relevant for L-shaped Indifference curves? (1+1+1)
3. Define inflation in an economy. State the differences between cost-push and demand-pull inflation. (1+2)
4. What is the purpose of fixing a base year for calculating the GDP of a country? (2)
5. A student derives utility from two goods: books and movies. The utility function of the student is represented as: $u(\text{books}, \text{movies}) = \sqrt{\text{books} \times \text{movies}}$. The student has a total budget of 40 rupees to spend, and the prices of the goods are as follows:
 - The price of one book is 20 rupees.
 - The price of one movie ticket is 10 rupees.

Using the Lagrangian multiplier method, determine the number of books and movie tickets the student should purchase to maximize his utility, subject to the budget constraint. (6)

OR

6. Explain in graphically how, in a Keynesian framework, an autonomous increase in government expenditure influences the equilibrium income in the economy. Does an increase in taxes have the same impact on equilibrium income? Explain. (4+2)

2nd Half

Module III: Management (Marks-16)

(Attempt any 2 questions)

1. Eco Gear, a startup founded in 2023, aimed to revolutionize the athletic wear market with their sustainable, eco-friendly workout clothing line. The company used recycled materials and environmentally conscious manufacturing processes to create high-performance athletic wear. Despite having an innovative product line, Eco Gear faced significant challenges in:
 - Building brand awareness in a saturated market
 - Convincing customers to pay premium prices for sustainable products

Indian Institute of Engineering Science and Technology, Shibpur
BTech/BArch 7th Semester End-Semester Examination, November, 2024

Finance Economics and Management for Engineers (HU4101)

Full Marks: 50

Time: 3 hours

- Establishing effective distribution channels
- Creating a marketing mix that would resonate with environmentally conscious consumers

Eco Gear developed a comprehensive marketing strategy which are elucidated below:

a. Product

- Developed a signature fabric made from recycled ocean plastics
- Created minimalist, versatile designs
- Used eco-friendly packaging
- Offered a product recycling program

b. Price

- Implemented a premium pricing strategy
- Introduced a loyalty program offering discounts on recurring purchases
- Created bundle pricing for workout sets

c. Place

- Launched a D2C e-commerce website
- Partnered with select high-end fitness studios
- Opened pop-up stores in environmentally conscious communities

d. Promotion

- Leveraged social media influencers in the fitness and sustainability space
- Implemented a content marketing strategy focusing on environmental impact
- Created a "Green Miles" rewards program
- Organized beach cleanup events with customer participation

After six months the startups achieved the following results:

- Achieved 150% of projected sales targets
- Gained 100,000+ social media followers
- Diverted 50,000 plastic bottles from oceans
- Established partnerships with 25 fitness studios

Questions:

[4 x 2 = 8]

- i. Which elements of the marketing mix appear to have contributed most to their success?
 - ii. Identify Eco Gear's target market segments. How could they expand their target market while maintaining their brand identity?
 - iii. Suggest additional distribution channels that could benefit the company.
 - iv. What recommendations would you make for long-term brand development?
-

2. Consider a scenario where a new CEO takes over a company that has faced multiple ethical scandals under previous leadership. The CEO is committed to restoring the company's ethical reputation and creating a transparent organizational culture.

Questions:

[2 x 4 = 8]

- i. What specific actions should the CEO take to promote an ethical culture?
 - ii. How can ethical leadership positively impact employees and stakeholders?
-

3. Case Study: Management Styles in Indian Manufacturing

Consider a highly structured manufacturing plant in India where a production manager follows an **autocratic management style**. The manager makes all decisions independently, sets strict production targets, and expects employees to follow instructions without much room for feedback or flexibility. This approach has helped the plant meet tight deadlines and maintain high output. However, employees report feeling unmotivated and undervalued due to the lack of participation in decision-making. Turnover rates are higher than industry standards, and the quality of work is beginning to show signs of inconsistency.

Based on the autocratic management style described, analyze the possible benefits and drawbacks of this leadership approach in a manufacturing environment. As an engineering student, discuss how an alternative management style, such as **democratic** or **laissez-faire**, could potentially impact productivity, employee morale, and overall efficiency in such a setting.

In your response, consider:

- i. The impact of autocratic management on employee motivation and retention. (2)
 - ii. How involving employees in decision-making could influence productivity. (2)
 - iii. The suitability of different management styles for a manufacturing setup from an engineering perspective. (4)
-

4. Case Study: International Strategies of Indian Multinational Companies

Sun Pharmaceuticals operates in countries like the U.S., Russia, and Brazil, using a **transnational strategy** that balances cost efficiency with local responsiveness. The company offers standardized products for economies of scale but customizes drug formulations and packaging to meet local health regulations and preferences.

Evaluate Sun Pharma's transnational strategy. As an engineering student, discuss how balancing standardization and customization might impact the company's operational efficiency and regulatory compliance. Consider:

- i. Advantages of transnational strategy for international operations. (2)
- ii. Challenges of product customization in foreign markets. (2)
- iii. Effects on manufacturing processes and compliance requirements. (4)

Module I

FINANCE

Answer any one question. Each question carries 10 marks.

1.a Define Cost. b. What is Costing? c. Briefly explain the Methods of costing. d. Why different Methods of costing are needed? (2+1+5+2)

2.a.Explain cost classifications based on traceability with examples.

b. Calculate Selling & Distribution overheads:

Power & Fuel ₹10,000/

Advertisement ₹1,80,000/

Meeting expenses ₹54,000/

Salesmen's Salaries & Commissions ₹1,00,0000/

Packing Charges ₹56,000/

Debtors ₹1,34,00/

Creditors ₹32,000/

Carriage outwards ₹67,000/

Showroom Rent ₹70,000/

Office Rent ₹95,000/

Show Room Lighting ₹58,000/

Sales ₹5,32,000/

Stores Manager' Salary ₹80,000/

(2+8)

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Module II

ECONOMICS

Attempt any one part.

PART-A

1. (a) Define and state the differences between GDP and GNP.

(b) Suppose you sell your three year old mobile phone to your friend. Will the income earned by you be counted in the current year's GDP calculation? Explain your answer. (2+1)

2. Complete the statements taking the correct option from the options provided below :

(a) In the circular flow of income model the factor payments/wages are made/given to

.....

(i) Firms (ii) Households (iii) Foreign Firms (iv) all of the above.

(b) In the circular flow of income model, without leakages and injections, the magnitude of income flowing in the system remains.....

- (i) Constant (ii) increases as time passes (iii) undecided (iv) doubles as time passes
(1+1)

3. Assume a country named "X" produces only two goods viz. Apples and Oranges. Suppose in the year 2023 price of apples is Rs. 20 and price of oranges is Rs. 12 while their production is 100 kgs and 150 kgs respectively in the country. Calculate the Nominal GDP of "X" for the year 2023 and also the Real GDP for the year 2023 with base year as 2020. Price of apples and oranges in 2020 were Rs.5 per kg and Rs.8 per kg respectively.
Define GDP deflator. (1+1)

4. Calculate GDP and GNP. Also Calculate GNP at Factor Cost:
Final Consumption expenditure ₹ 300; Domestic Capital formation (Investment) ₹800;
Exports ₹250; Imports ₹550; Expenditure of the government ₹200; Payments to factors abroad ₹100; Income from abroad ₹500; Net indirect tax ₹100; Debt interest income ₹20; Depreciation ₹25. (3)

PART-B

5. State the functions of money. State which functions of money the following items satisfy. a. Cash money b. Bitcoin c. Real Estate property (1+3)
6. State the differences between Commercial Bank and a Central Bank of a country. Explain how commercial banks create money supply. (2+4)

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Module III MANAGEMENT

Answer any two questions. Each question carries 5 marks.

1. Illustrate diagrammatically and define the different types of managerial styles as prescribed by the managerial grid. (2.5+2.5)
2. Write a brief note on the Hawthorne studies. (5)
3. What are the different types of managerial functions? (5)
4. Differentiate between transactional and transformational leadership styles. (2.5+2.5)

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02/05/2023

Indian Institute of Engineering Science and Technology , Shibpur

B.Tech 4th Year 8th Semester(CE/ME/Min) Final Examination, May '23

FINANCE ECONOMICS & MANAGEMENT for ENGINEERS(HSS III)(HU - 4201)

(Use Separate Answer scripts for two halves. Answer Modulewise and start writing

Answers for each Module on a separate page)Full Marks : 50Time : 3 hoursEach Module carries 16 marks (3x16=48). 2 marks for neatness1st HalfModule IA Explain with an imaginary example the concept of (any Two)

i. Operating Leverage ii. Financial Leverage iii. Combined Leverage.

(2x4)

B. i. Explain what is NPV Or PBP Or IRR.

ii. Give your opinion based on NPV regarding the acceptability of the following Two projects with reason: .

	<u>Project A</u>	<u>Project B</u>
Initial Outlay	Rs 1,00,000	Rs 1,00,000
Cost of Capital	10%	10%
Estimated Inflows Year 1	Rs 10,000	18,000
2	25,000,	27,000
3	35,000	36,000
4	47,000	49,000
5	10,000	2,000(2+6)
		(See Table if needed)

Module IIEconomics

Answer both question No 1&2

1. Use the following information to answer questions a to d:

The table below shows data for Apples for an individual firm.

Quantity of Apples	Fixed Costs (FC)	Variable Costs (VC)	Total Costs (TC)
0	?	?	80
1	80	117	197

2	80	252	332
3	?	?	485
4	80	?	656
5	80	765	845
6	80	?	1052
7	?	1197	1277
8	80	1440	1520

- Given this data, what are fixed costs when quantity is 0?
- Given this data, what are variable costs when quantity is 0?
- Given this data, what are variable costs when quantity is 3?
- Given this data, what are variable costs when quantity is 4? (4)

- Assume that a manufacturing company produces 1000 units and sells them at a price of Rs 5 each. The Average Total Cost (ATC) is Rs 7,000 with a fixed cost (FC) of Rs 4000 and a variable cost (VC) of Rs 3,000 for all units. Evidently, this manufacturing company is operating at a loss of Rs 2000 (economic loss). Whether this firm should continue its production in the short run? Give proper reasons. (5)
 - What is planning curve? (2)

Attempt any one question from below :

- How to determine National Income by Income Method? (5)
- What is Investment Multiplier? Why the theory of multiplier does not generally apply to Less Developed Countries? (5)

2nd Half

Module III

Management

Answer any two questions. Each question carries 8 marks.

- Differentiate between strategic and operational planning. Explain the concept of differentiation strategy with suitable examples. (4+4)
- What conditions encourage a firm to pursue stability strategies? What are the different types of expansion strategies? (4+4)
- What are the advantages of the managerial grid? Discuss the advantages and disadvantages of the different channels for transmitting a message. (2+6)

Table C: Present Value Factor of a Lump Sum (PVF) of Re 1

Year	Interest Rate													
	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	11%	12%	13%	14%
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	0.901	0.893	0.885	0.877
2	0.980	0.961	0.943	0.925	0.907	0.890	0.873	0.857	0.842	0.826	0.812	0.797	0.783	0.769
3	0.971	0.942	0.915	0.889	0.864	0.840	0.816	0.794	0.772	0.751	0.731	0.712	0.693	0.675
4	0.961	0.924	0.888	0.855	0.823	0.792	0.763	0.735	0.708	0.683	0.659	0.636	0.613	0.592
5	0.951	0.906	0.863	0.822	0.784	0.747	0.713	0.681	0.650	0.621	0.593	0.567	0.543	0.519
6	0.942	0.888	0.837	0.790	0.746	0.705	0.666	0.630	0.596	0.564	0.535	0.507	0.480	0.456
7	0.933	0.871	0.813	0.760	0.711	0.665	0.623	0.583	0.547	0.513	0.482	0.452	0.425	0.400
8	0.923	0.853	0.789	0.731	0.677	0.627	0.582	0.540	0.502	0.467	0.434	0.404	0.376	0.351
9	0.914	0.837	0.766	0.703	0.645	0.592	0.544	0.500	0.460	0.424	0.391	0.361	0.333	0.308
10	0.905	0.820	0.744	0.676	0.614	0.558	0.508	0.463	0.422	0.386	0.352	0.322	0.295	0.270
11	0.896	0.804	0.722	0.650	0.585	0.527	0.475	0.429	0.388	0.350	0.317	0.287	0.261	0.237
12	0.887	0.788	0.701	0.625	0.557	0.497	0.444	0.397	0.356	0.319	0.286	0.257	0.231	0.208
13	0.879	0.773	0.681	0.601	0.530	0.469	0.415	0.368	0.326	0.290	0.258	0.229	0.204	0.182
14	0.870	0.758	0.661	0.577	0.505	0.442	0.388	0.340	0.299	0.263	0.232	0.205	0.181	0.160
15	0.861	0.743	0.642	0.555	0.481	0.417	0.362	0.315	0.275	0.239	0.209	0.183	0.160	0.140
16	0.853	0.728	0.623	0.534	0.458	0.394	0.339	0.292	0.252	0.218	0.188	0.163	0.141	0.123
17	0.844	0.714	0.605	0.513	0.436	0.371	0.317	0.270	0.231	0.198	0.170	0.146	0.125	0.108
18	0.836	0.700	0.587	0.494	0.416	0.350	0.296	0.250	0.212	0.180	0.153	0.130	0.111	0.095
19	0.828	0.686	0.570	0.475	0.396	0.331	0.277	0.232	0.194	0.164	0.138	0.116	0.098	0.083
20	0.820	0.673	0.554	0.456	0.377	0.312	0.258	0.215	0.178	0.149	0.124	0.104	0.087	0.073
25	0.780	0.610	0.478	0.375	0.295	0.233	0.184	0.146	0.116	0.092	0.074	0.059	0.047	0.038
30	0.742	0.552	0.412	0.308	0.231	0.174	0.131	0.099	0.075	0.057	0.044	0.033	0.026	0.020
40	0.672	0.453	0.307	0.208	0.142	0.097	0.067	0.046	0.032	0.022	0.015	0.011	0.008	0.005
50	0.608	0.372	0.228	0.141	0.087	0.054	0.034	0.021	0.013	0.009	0.005	0.003	0.002	0.001

Indian Institute of Engineering Science and Technology, Shibpur
B Tech/B Arch 7th Semester End-Semester Examination, November, 2023
FINANCE ECONOMICS & MANAGEMENT for ENGINEERS(HSS III)(HU - 4101)

Full Marks:50

Time: 3 Hours

(Use Separate Answer scripts for two halves. Answer Module wise and start writing Answers for each

Module on a separate page)

Each Module carries 16 marks (3×16=48) + 2 marks for neatness

1st Half

Finance Module

1. Explain with example Element wise OR Behaviour wise Cost classification. (6)
2. Explain briefly the long Term Sources of Finance. (4)
3. What is NPV? From the following Information give your opinion about which project should be selected assuming the discounting rate being 10% ?

Name of the Project	<u>Project A</u>	<u>Project B</u>
Life	4 Years	5 Years
Investment	₹2Lakhs	.5 Lakhs
Estimated Cash Inflows(₹)		
Year 1	50,000	35,000
2	56,000	66,000
3	61,000	64,000
4	72000	68000
5	nil	23,000

PV Factor at 10% for 1 to 5 years are 0.909/0.826/0.751/0.683/o.621(2+4)

Management Module

Answer any two questions. Each question carries 8 marks.

4. List the 4 phases of the strategic management process. According to the Porter's 5 forces model what are the possible threats of entry to a new entrant in an industry? (2+6)
5. Illustrate diagrammatically and write a short note on the Ansoff matrix. How can a company achieve cost leadership? (5+3)
6. Give a detailed explanation of the different linear models of communication. (8)

2nd Half

Economics Module

7. Explain simple Keynesian model of Income Determination graphically. Explain the concept of Investment multiplier. Assume the value of marginal propensity to consume rises, what will be its effect on the magnitude of investment multiplier? (3+1+2)
8. State the law of demand. Keeping in mind that price is determined at the point of interaction of supply and demand curves, consider a situation when there is a rise in income of the individual. What will happen to the equilibrium price? Explain using demand and supply curves. (1+4)
9. What is the law of diminishing marginal utility? Derive the demand curve using the concept of diminishing marginal utility. (1+4)

OR

10. Define GDP. Refer to the table below and calculate the value of GNP. Statement: Rich countries have high GDP. Hence high GDP of a country ensures greater welfare of the people. What is the problem with the above statement? (1+2+2)

<u>Particulars</u>	<u>Dollars(\$)</u>
Final Consumption expenditure	450
Investment	700
Exports	1000
Imports	550
Expenditure of the government	300
Payments to factors abroad	200
Income from abroad	500
Net indirect tax	100
Debt interest income	20
Depreciation	25

B.Tech/ B Arch 4th Year 7th Semester(AE/MET/CST/ETC/IT/EE/ARCH)

Mid Term Examination Sept'22

FINANCE ECONOMICS & MANAGEMENT for Engineers(HSS III)(HU - 4101)

Full Marks:30 (Start writing Answers for each Module from a separate page) Time:2hours

Module I

Finance

I. What do you understand by the Term Cost? (2)

II . Explain Element wise classification of cost
Or Functional Classification cost. (3)

III . From the given data, calculate only Works Overheads :

Items	Rs
Direct Materials	10,00,000/
Direct Wages	2,00,000/
Indirect Materials	60,000/
Indirect Wages	1,00,000/
Advertisement	50,000/
Rent:	
Office	1,15,000/
Factory	75,0000/
Lighting	88,0000/
(50% for Factory)	
Power	60,000/
Depreciation of Plant	1,15,500/
Salary of Works Manager	95,000/
Lubricant	10,000/
Factory Guards Salary	30,000/
Salary of Administrative staff	80,000/
Insurance of Plant	25,000/

(5)

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Module II
ECONOMICS

A. The supply and demand functions are given below. Determine which one among these equations is the equation of supply and which one is the equation of demand and explain with logic. Find the equilibrium price (in Rs) and quantity (in Kgs). (4)

$$P = -2Q + 100 \text{ and } P = 3Q - 30$$

B. What are the assumptions of cardinal utility analysis? (3)

C. As a result of a 10% increase in the income of a consumer, its demand rises from 200 to 240 units. Find out the income elasticity of demand and comment on the nature of goods. (3)

OR (this alternative is only for question C)

What are the characteristics of the Indifference Curve and what do you mean by budget line? (2+1=3)

#####

Module III
Management

Answer any two questions. Each question carries 5 marks. (2 × 5 = 10)

1. What is Management? Discuss the features or characteristics of management.
2. Discuss different levels of Management.
3. Discuss 'Management is both a science as well as an art'.

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28/11/22

B.Tech/ B Arch 4th Year 7th Semester(AE/MET/CST/ETC/IT/EE/ARCH)
END Semester Examination November, 2022

FINANCE ECONOMICS & MANAGEMENT for Engineers (HSS III) (HU - 4101)
Answer Module I & II in the 1st Half and Module III in the 2nd half answer scripts.

FullMarks:50

Two marks for neatness .

Time: 3 hours

Module I
Finance (16)

A. From the following information prepare a Cost Sheet for the month ended 31st October 2022:

<u>Items</u>	<u>Amount (Rs.)</u>	
Raw material :	1,50,000	
Direct Labour :	1,00,000	
Packaging charge:	10,000	
Lubricant :	5,000	
Office salary :	24,000	
Factory Rent :	10,000	
Postage :	500	
Salesman salary :	10,000	
Sales :	3,50,000	[8]

B. What is Leverage? Explain with example the concept of different types of Leverages. [8]
OR

Discuss the different sources of Long Term Funds available for Corporate Finance. Also discuss the concept of Cost of Capital in this context. [8]

Module II
ECONOMICS (16)

- (a) State and explain the Law of Variable Proportions with a diagram.
(b) What is the relationship between the average product and marginal product curve? (4+2)
- (a) Why the minimum point of the short-run average cost (SRAC) curve lies to the right of the minimum point of the short-run marginal cost (SRMC) curve? Why SRAC curve is U shaped?
(b) What are the differences between Isoquant and Iso-cost? How does a producer achieve equilibrium?
(c) What is the difference between opportunity and sunk cost? (3+2+1)
- (a) "All marketable production is economic production but all economic production is not marketable"- Explain.
(b) When it will have a contractionary effect on the circular flow of the economy of a three-sector model? What are the leakages and injections in the economy of four sector model? (2+2)

Or (this alternative is only for question 3)

- (a) What is an investment multiplier?
-

(b) Suppose the nominal GNP of a country i.e. GNP estimated at current prices, in the year 2012 is given at INR 500 billion, and the price index number is given as base year 2012=100. Now let the nominal GNP increase to INR 600 billion in the year 2017 and PIN rise to 110. Find the real GNP. (2+2)

Module III

Management (16)

Answer any FOUR (4 x 4=16)

- I. Explain three basic styles of Management.
 - II. Discuss Ansoff Product-Market Matrix.
 - III. Under what conditions would you recommend to use Turn Around Strategy in an organization?
 - IV. Distinguish between Concentric and Conglomerate Diversification with suitable examples.
 - V. Differentiate between Cost Leadership and Differentiation Strategies.
 - VI. Explain the process of Communication.
-